

Illini
Electric
Motorsports

DAQA Circuits IDR

10/15/27

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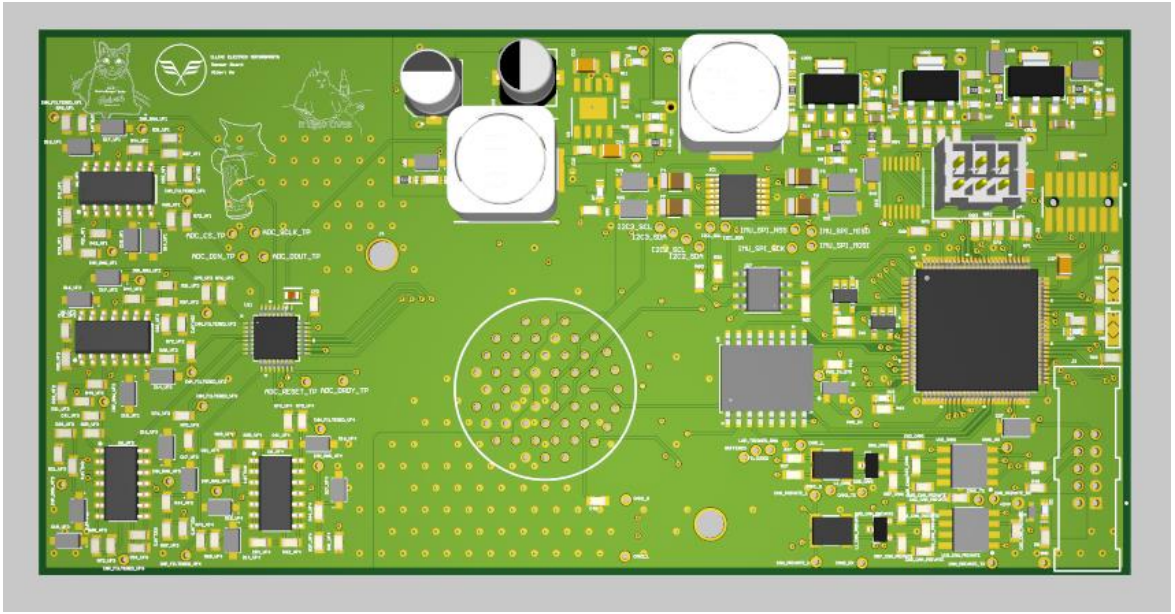
Agenda

- Board Purpose
- Roses/Thorns
- Current Board Overview
- ADC Selection
- Analog Signal Filtering Chain
- Sensor Powering OCP
- Layout Changes
- Conclusion

Board Overview

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- **Data Acquisition (DAQA)/Sensor Board**
- Provides OCP and power to non car critical sensors throughout the car
- Collects and processes digital and analog sensor data
- Transmits sensor data over CAN for logging and analysis
- Cannot be car critical == nothing is rules-required/drive-required



Last years board for aura

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Roses/Thorns

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Roses

- Layout and schematics were nice and well organized
- Good software and CD integration
- Digital sensor support
- Better analog implementation

Thorns

- Component Selection Issues
  - ADC input
- Off board power buffers did not work and got really hot
- Not enough time to test DAQA board
  - Sidelined for car critical stuff
- Expensive board + autosport
- Excessive bus load contribution (config-dependent)
- CPU utilization/jitter/timing issues

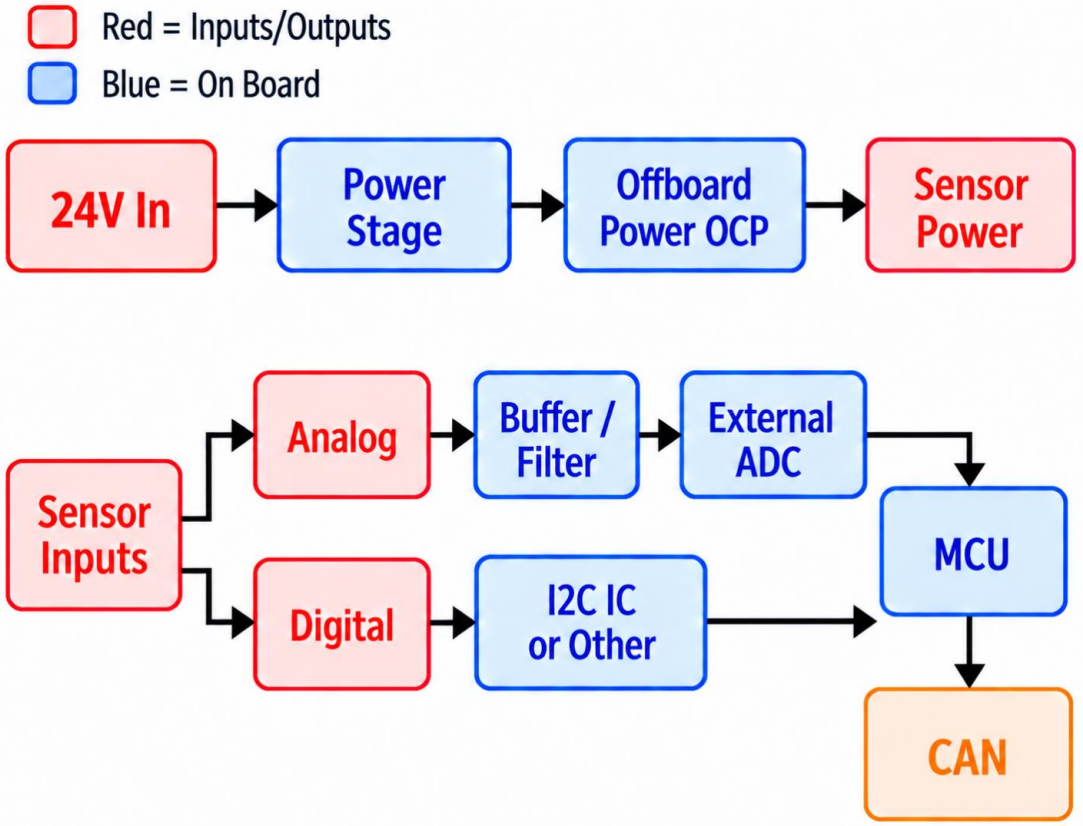
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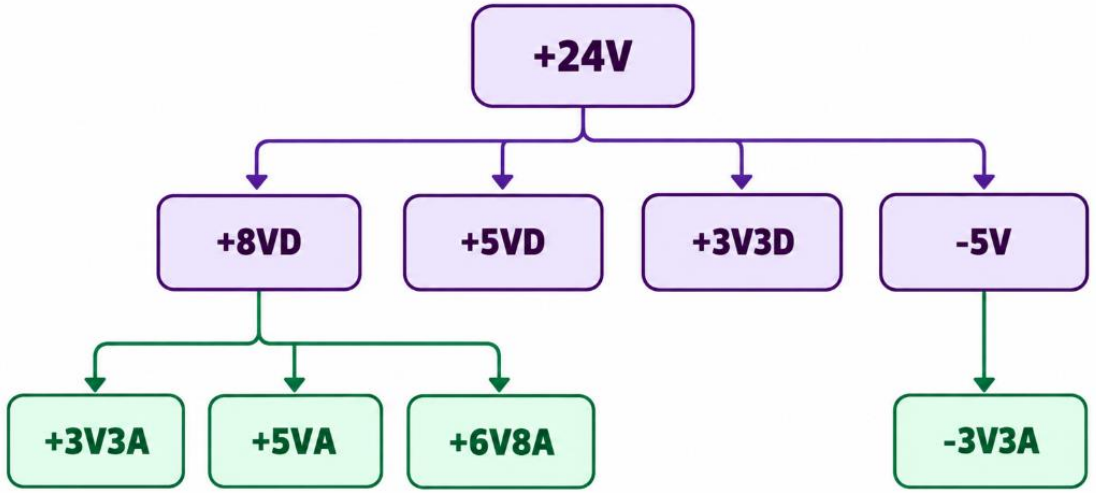
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Board Overview

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3V3D, 5VD, 5VA offboarded
Purple = bucks
Green = LDOs

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Trade #1 – ADC Selection

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Question:

Will a higher performance ADC improve system performance without introducing unnecessary trade-offs?

DCR Requirements:

- **PFR.3:** Sensor board shall support the appropriate connectors and communication protocols (private CAN, SPI, I2C, etc.) for all sensors
- **PFR.5:** Logging rates and accuracy for individual sensors shall be sufficient to meet the objectives defined by the sensor's requesting subsystem or stakeholder(s).
- **PFR.9:** Sensor board software shall be implemented as a single codebase compatible with all planned hardware and sensor configurations.



Image shown is a representation only. Exact specifications should be obtained from the product data sheet.

ADS131M08QPBSRQ1

DigiKey Part Number	296-ADS131M08QPBSRQ1TR-ND - Tape & Reel (TR) 296-ADS131M08QPBSRQ1CT-ND - Cut Tape (CT) 296-ADS131M08QPBSRQ1DKR-ND - Digi-Reel®
Manufacturer	Texas Instruments
Manufacturer Product Number	ADS131M08QPBSRQ1
Description	IC ADC 24BIT 32-TQFP
Manufacturer Standard Lead Time	26 Weeks
Customer Reference	
Detailed Description	24 Bit Analog to Digital Converter 8 Input 8 Sigma-Delta 32-TQFP (5x5)
Datasheet	Datasheet

2025-2026 ADC

Requirements:

8 inputs enough

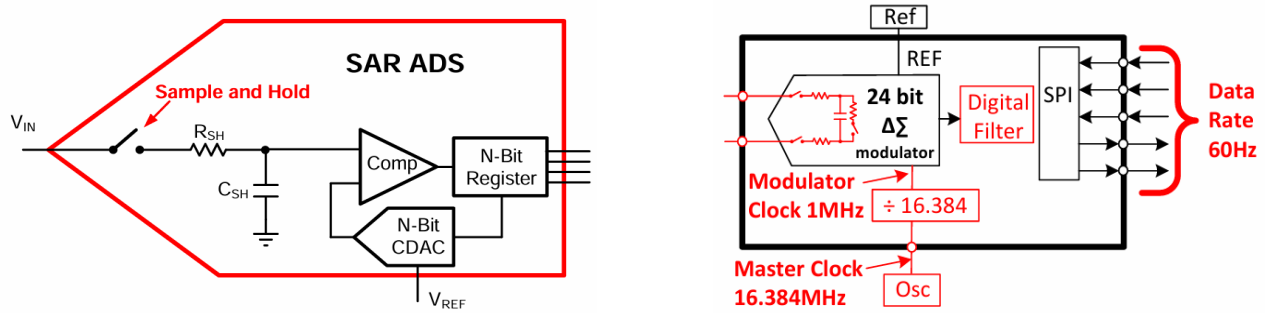
24 bits nominal enough

16.8 bits ENOB enough

Trade #1 – ADC Selection

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- **Architecture:** DS vs SAR
- **Factors:**
  - Difficulties of implementing a new ADC
  - Effective resolution needed
  - Power consumption
  - Number of channels
  - Input Characteristics (Single-ended/Differential)



SAR vs. DS Diagrams

- Current ADC chosen due to range, single-ended capabilities, smaller footprint.

Trade #1 – ADC Selection

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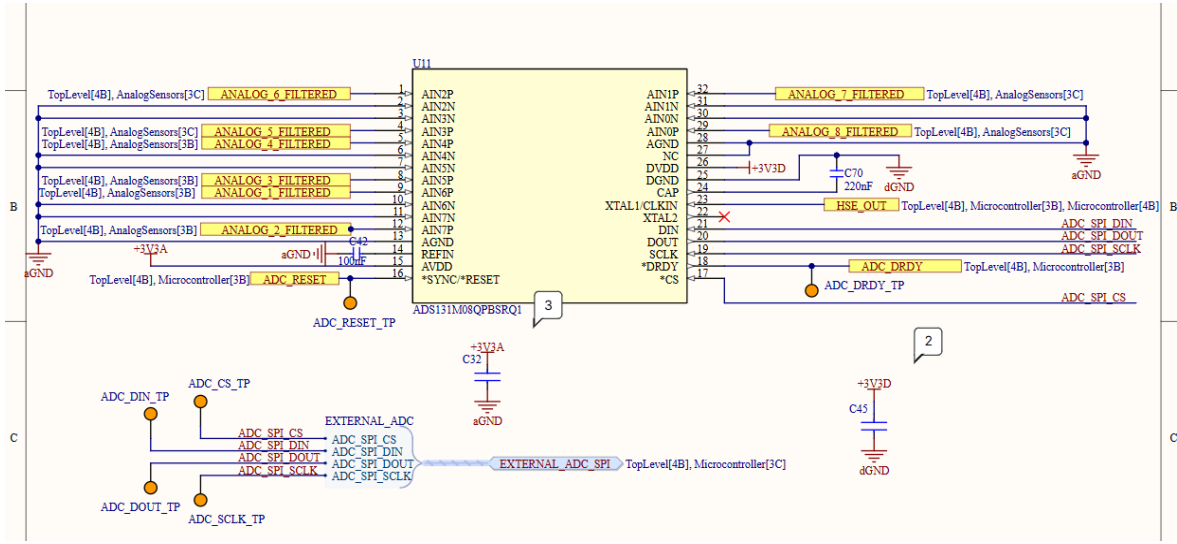
Category	Option #1: ADS8688AIDBTR SAR	Option #2: ADS131M08QPBSRQ1 (Current) DS	Option #3: ADS131M06QPBSRQ1 and ADS131M04QPWRQ1 DS	Option #4: AD7124-8WBBCPZ-RL7 DS
Power Consumption	Highest power consumption 65mW	Low power consumption 6mW	Lowest power consumption 4.9mW / 3.3 mW	Low power consumption 7.1mW
Implementation Difficulty	Needs new firmware, longer time to implement	No change – drivers already written	Existing driver supports all ADS131M0x variants (4 or 6 channels)	Needs new firmware, longer time to implement, extra channels
Resolution & Noise	16-bit, highest sample rate	24-bit 16.8 bits ENOB @ 32kSPS / gain: 1	24-bit 16.8 bits ENOB	24-bit 16.1 bits ENOB @ 19.2kSPS / gain: 1
Cost	$23.06/u	$12.50/u	$11.11/u and $8.96/u	$23.73/u

Trade #1: Selection

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Selection: (ADS131M08QPBSRQ1)

- We are choosing to keep our current ADC
  - Eliminates need for firmware change
  - Voltage input range: +-1.2V
  - 8 channels
  - Cost-effective option
  - Effective resolution sufficient
  - Power efficient (6mW)



Implementation

- Same as last year

2026 Sensor Board V2 ADCs

Table 7-2. Dynamic Range (Effective Resolution) at T_A = 25°C

OSR	DATA RATE (kSPS), f_{CLKIN} = 8.192 MHz	GAIN							
		1	2	4	8	16	32	64	128
16384	0.25	113 (20.3)	108 (19.4)	103 (18.6)	101 (18.3)	98 (17.8)	96 (17.5)	90 (16.5)	84 (15.4)
8192	0.5	111 (19.9)	106 (19.1)	100 (18.1)	98 (17.8)	96 (17.4)	93 (17.0)	87 (16.0)	81 (15.0)
4096	1	108 (19.4)	103 (18.6)	97 (17.7)	96 (17.4)	93 (17.0)	91 (16.6)	85 (15.6)	79 (14.6)
2048	2	106 (19.1)	101 (18.2)	95 (17.3)	93 (17.0)	91 (16.6)	88 (16.2)	82 (15.2)	76 (14.2)
1024	4	104 (18.8)	99 (18.0)	93 (17.0)	92 (16.8)	89 (16.3)	87 (15.9)	81 (14.9)	75 (13.9)
512	8	101 (18.3)	96 (17.5)	90 (16.5)	89 (16.3)	86 (15.8)	84 (15.4)	78 (14.4)	72 (13.4)
256	16	98 (17.8)	93 (16.9)	87 (16.0)	86 (15.8)	83 (15.3)	81 (14.9)	75 (13.9)	69 (12.9)
128	32	92 (16.8)	89 (16.3)	84 (15.4)	83 (15.2)	80 (14.8)	78 (14.4)	72 (13.4)	65 (12.4)

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Trade #2 – Analog Signal Processing

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- **Problem:** each analog sensor has its own output impedance, and the sensor, RC filter, divider, and ADC input all load each other, shifting readings and filter cutoffs
- **Question:** where do we need op-amp buffers? Is last year's double op-amp setup needed, or can we drop one to save board space and power?
- **Last Year:** sensor → op-amp → RC filter → op-amp → ADC
- **DCR:** PFR.4 accurate, usable ADC measurements; PFR.6 support sensor output ranges

Options

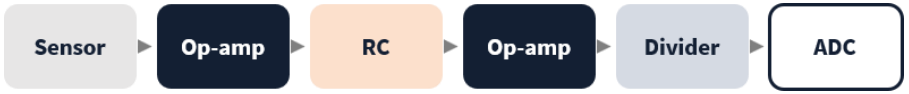
#1 No Buffer



#2 Sensor-side buffer only

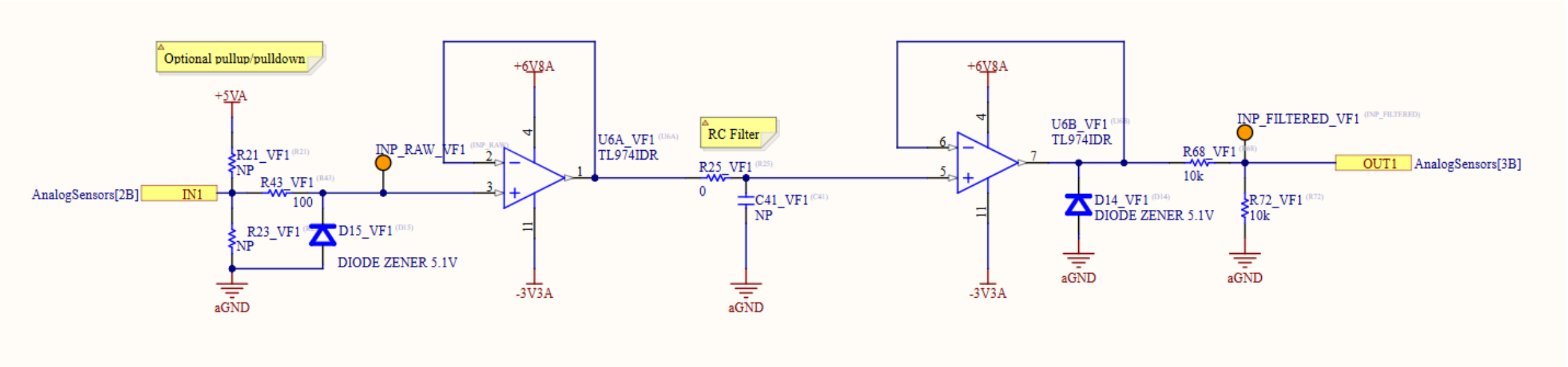


#3 Sensor + ADC-side buffers



Trade #2 – Last Year

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2026_Sensor_Board_V2: OpAmpVoltageFollower (VF1)

Last year

- **Optional pullup/pulldown:** for resistive or switch sensors (e.g. thermistor divider), used when needed
- **Sensor-side buffer** (U6A)
- **RC filter** between the followers (values set per sensor)
- **ADC-side buffer** (U6B) drives the divider/protects filter
- **Resistor divider** → ADC

Compared to this year

- **Same architecture:** double buffer
- **RC filter:** sized with LTspice (e.g. 909 Ω + 220 nF, 796 Hz)
- **Divider:** same

Trade #2 – Sensor & ADC Impedances

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Sensors	Max Output Impedance
Pitot (PI-4)	47 Ω
Lin pot (MLS0950)	750 Ω
Load cell (TCM)	350 Ω
Thermistor (SEN-FM18T10)	10 kΩ at 25 °C

ADC	Max Input Impedance
ADC (ADS131M08)	330 kΩ

Sensor output impedance

Preferably lower, since it's in series with the signal; but our buffers draw almost no current, so it isn't a big issue

ADC input impedance

Higher means it draws less current, so it doesn't pull the signal's voltage down → higher is better

Examples used in our sims

Pitot: low, fixed impedance (47 Ω)
Lin pot: up to 750 Ω, changes with wiper position

Trade #2: Buffer Options

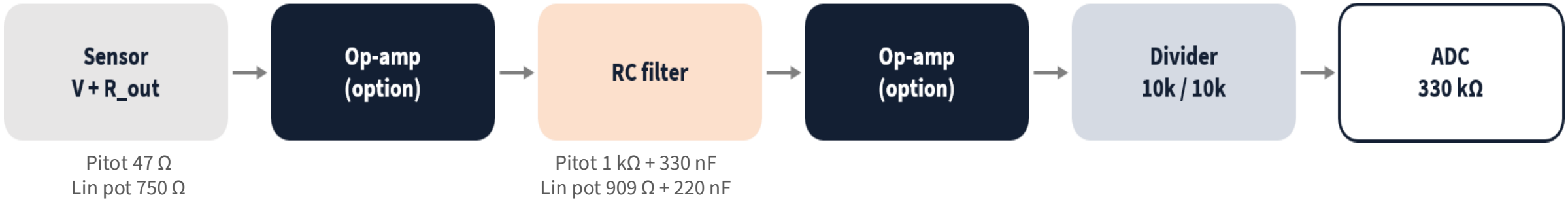
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Category	Option #1: No buffer	Option #2: Sensor-side buffer only	Option #3: Sensor + ADC-side buffers (chosen)
Sensor Impedance	Sensor's output impedance adds to the filter R, lowering the cutoff (and shifting it if the impedance changes during use)	Isolated from the filter	Isolated from the filter
Divider / ADC Loading	Sensor impedance + divider loading: signal 5-8% below ideal	Sensor impedance + divider loading: signal 4-5% below ideal	None: output hits ideal 0.4925, cutoff exactly $1/(2\pi RC)$
Lin pot sim result (designed: 0.4925, 796 Hz)	0.4543 / 473 Hz	0.4708 / 833 Hz	**0.4925 / 796 Hz**
Op-amps / board space	None	1 per channel (4 channels per TL974)	2 per channel (2 channels per TL974)

Trade #2: LTspice Setup

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Model



Frequency Domain

AC sweep, 1 Hz - 100 kHz
Measures: voltage fraction at the ADC and -3 dB cutoff

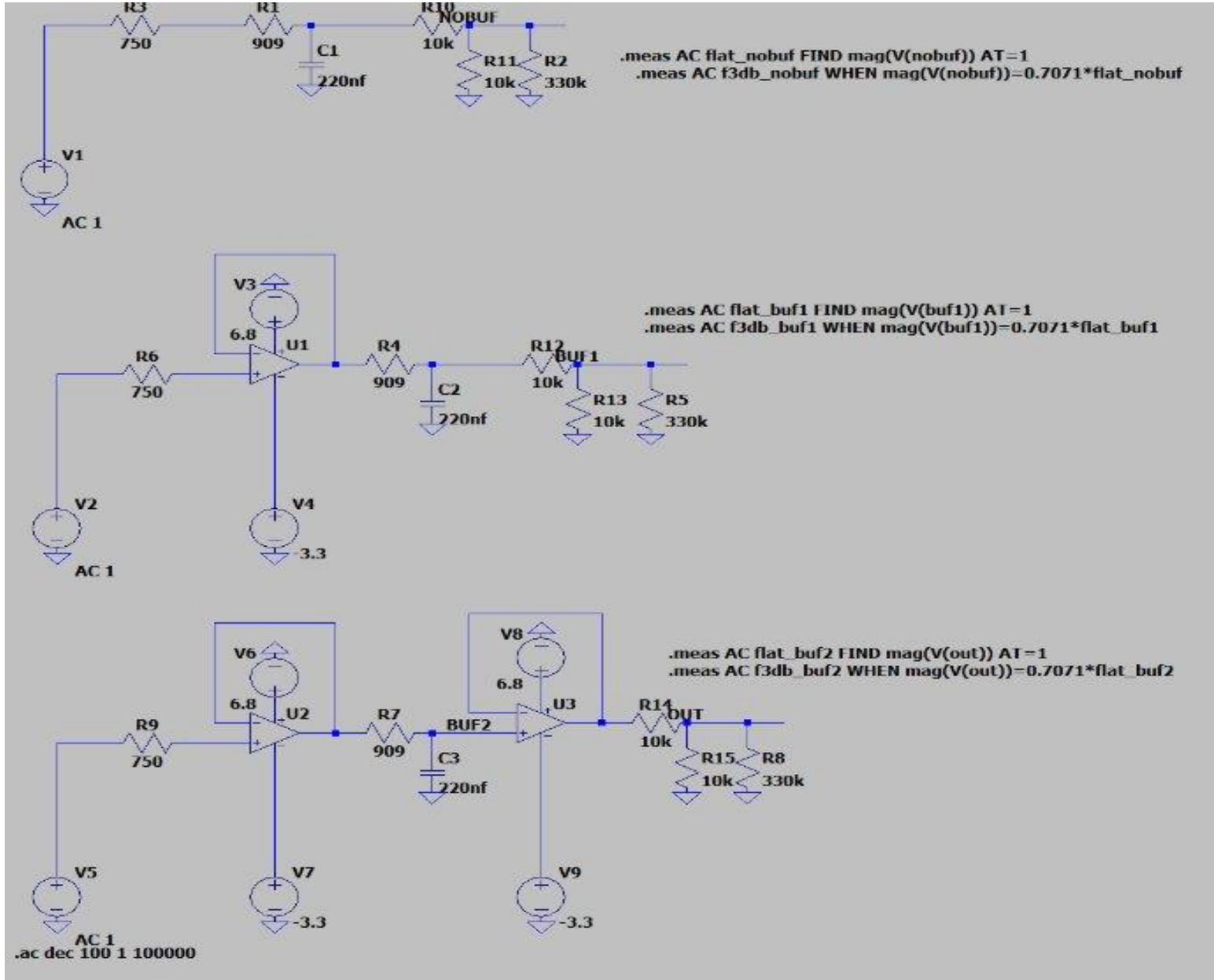
Time Domain

1 V pulse in
Measures: τ, t95, t99 = shortest real event the filter passes

Trade #2: LTspice Setup

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Lin pot model: no buffer / single / double



Trade #2: Frequency Domain - Pitot

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Config	Voltage fraction	-3 dB Cutoff
No Buffer	0.4677	485 Hz
Single Buffer	0.4687	507 Hz
Double Buffer	**0.4925**	**482 Hz**

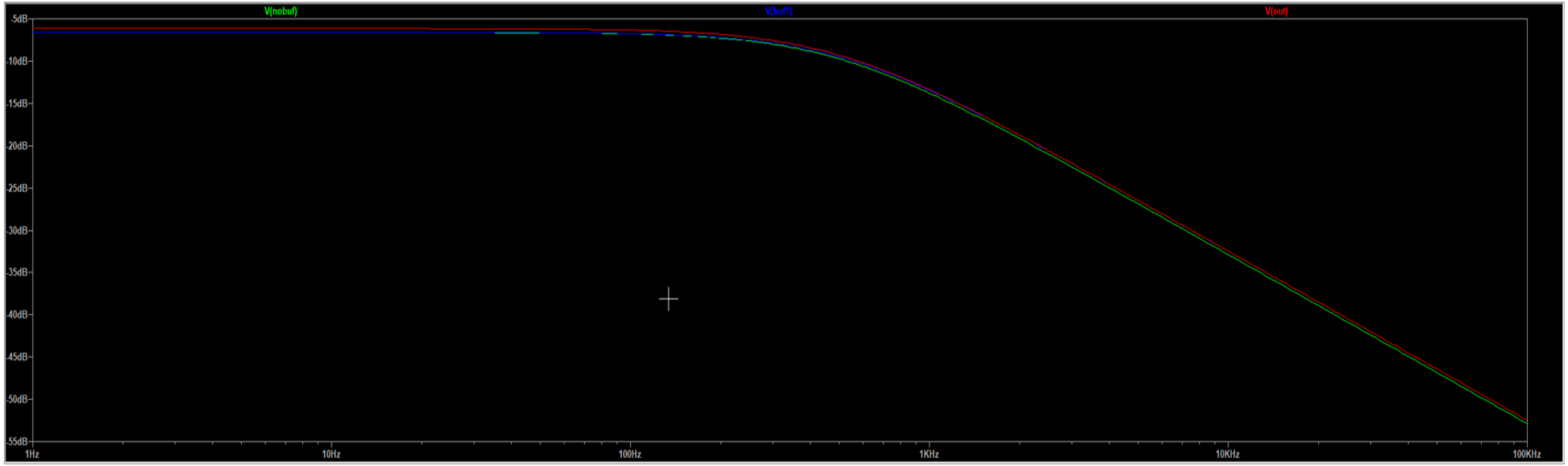
Designed: 0.4925 (divider only), 482 Hz

Takeaways

- **Low, fixed impedance:** parallel and series resistances barely moves the cutoff (485 vs 482 Hz)
- **Single:** divider loads the filter → ~5% low, cutoff up to 507 Hz
- **Double:** exact divider ratio and designed cutoff

Trade #2: Frequency Domain - Pitot

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Pitot Bode plot (1 Hz - 100 kHz): green = no buffer, blue = single, red = double

Trade #2: Frequency Domain - Linpot

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Config	Voltage fraction	-3 dB Cutoff
No Buffer	0.4543	473 Hz
Single Buffer	0.4708	833 Hz
Double Buffer	**0.4925**	**796 Hz**

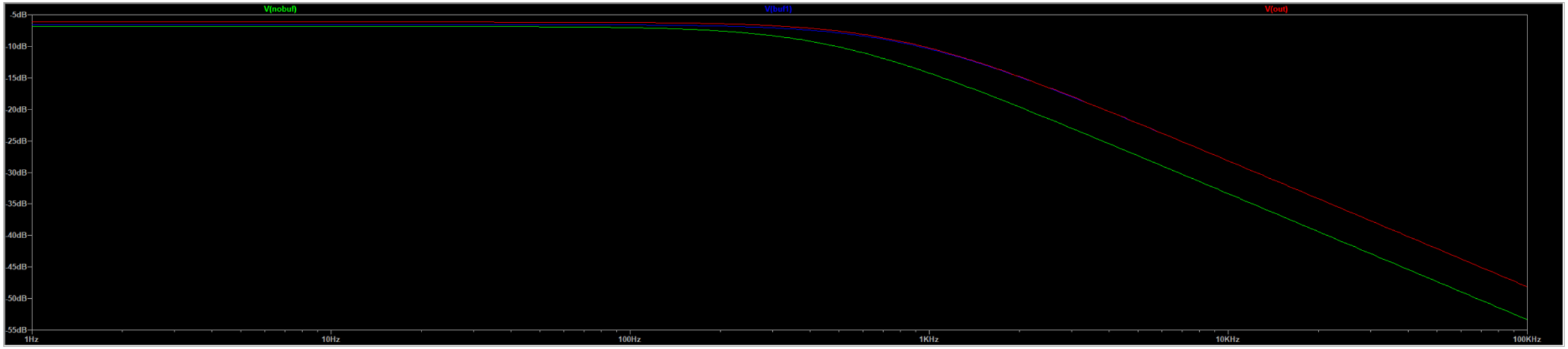
Designed: 0.4925, 796 Hz. Linpot at (750 Ω, mid-travel/worst case)

Takeaways

- **High, varying impedance:** unbuffered cutoff drops to 473 Hz at mid-travel and drifts as the wiper moves
- **Single:** divider shifts the cutoff to 833 Hz, ~4% low
- **Double:** exact divider ratio and essentially 800 Hz

Trade #2: Frequency Domain - Linpot

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Linpot Bode plot (1 Hz - 100 kHz): green = no buffer, blue = single, red = double

Trade #2: Time Domain

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1 V pulse in. $\tau = 63\%$, $t95 \approx 3\tau$, $t99 \approx 4.6\tau$ ($V = V_{final} \times (1 - e^{-t/\tau})$)

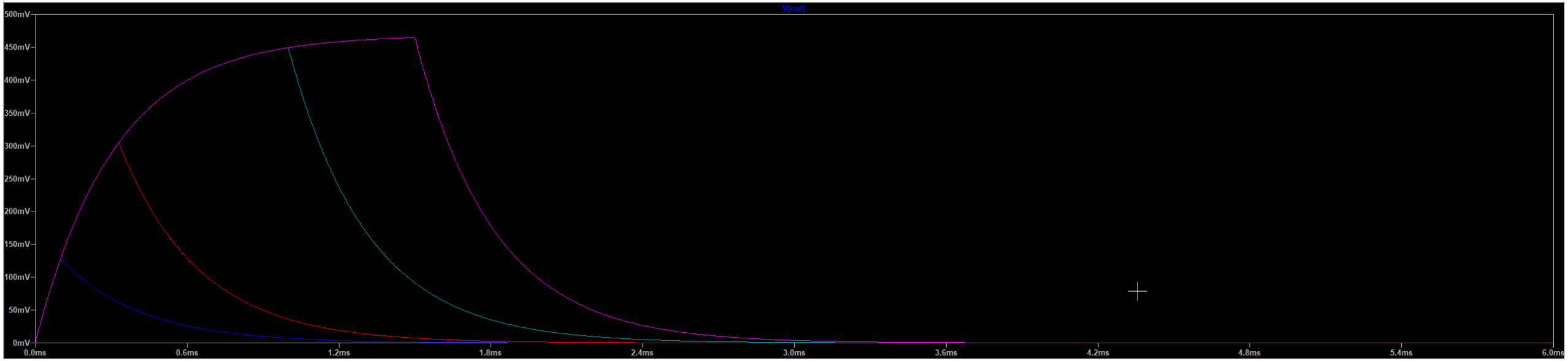
Sensor / Config	Final	τ	t95	t99
Pitot, single	0.4687	0.314 ms	0.941 ms	1.446 ms
Pitot, double	0.4925	0.330 ms	0.989 ms	1.520 ms
Load Cell / Linpot, single	0.4708	0.191 ms	0.573 ms	0.880 ms
Load Cell / Linpot, double	0.4925	0.200 ms	0.599 ms	0.921 ms

Takeaways

- **Shortest event at 95%:** ~1 ms (Pitot), ~0.6 ms (lin pot)
- **Confirmed with aero and vehicle dynamics:** real events are slower, so mostly only noise gets squashed
- **1 vs 2 buffers:** nearly the same timing

Trade #2: Time Domain

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Pitot output for 0.1 / 0.33 / 1 / 1.5 ms pulses: short spikes never reach full height

Trade #2: Load-Cell Interface

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- Wheatstone bridge amplification not needed for load cell signals
  - **ADC internal amplification**: for the current ADC, each channel has a programmable gain amplifier with settings from 1-128. Allows for increased sensitivity to smaller load-cell signals
  - **Internal strain-gauge bridge**: Load cell already produces differential voltage from force changes
    - Sensitivity ~2mV/V; differential output is 20mV at 10V excitation
    - ADC gain of 32 = 37.5mV of differential range, which means load cell output voltage in range
  - **High impedance ADC**: sensor voltage already preserved because of the ADC's high impedance

ADC Internal Gain

PGA Gain	Differential Range
32	37.5mV
64	18.75mV
128	9.375mV

Internal Reference: 1.2V
Range: 1.2V/gain

Trade #2: Selection

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Decision: double buffer for analog sensors. Single buffer for the coolant thermistor. Can decide to remove based on layout restrictions but more testing is needed on sensor by sensor basis if this was the case.

Analog sensors



Coolant thermistor



Why

- **Sensor-side buffer:** Isolates each sensor from the filter, so the sensor's output impedance can't pull down its own signal or shift the cutoff, giving more accurate readings. Required for linpots, whose datasheet specifies >500 kΩ wiper load; without a buffer, the filter + divider only load it at ≈ 20 kΩ, so a buffer is needed.
- **ADC-side buffer:** Removes divider/ADC loading, so the ADC reads exactly what we designed (the divider's exact ratio, cutoff = $1/(2\pi RC)$), with no extra sizing. Also lets each filter be sized independently of the divider and the ADC's PGA setting, which changes its input impedance
- **Thermistor:** Its divider is on the sensor side, so the filter is only loaded by the ADC's 330 kΩ (~0.3%). One buffer isolates the divider's changing impedance from the filter

Trade #2: Implementation

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- **Op-amp:** TL974IDR (4 op-amps per chip) → 2 sensors per chip with double buffering
- **Filters:** lin pot 909 Ω + 220 nF (~796 Hz, 800 Hz target); Pitot 1 kΩ + 330 nF (~482 Hz) **subject to change**
- **Validated:** time response (~0.6-1 ms) confirmed faster than real events by aero and vertical dynamics
- **Divider:** final ratios will be sized per sensor once the sensor list is set, so each max output lands just under the ADC's input voltage
- **Load cells:** read directly with the ADC's PGA gain (see load-cell slide)

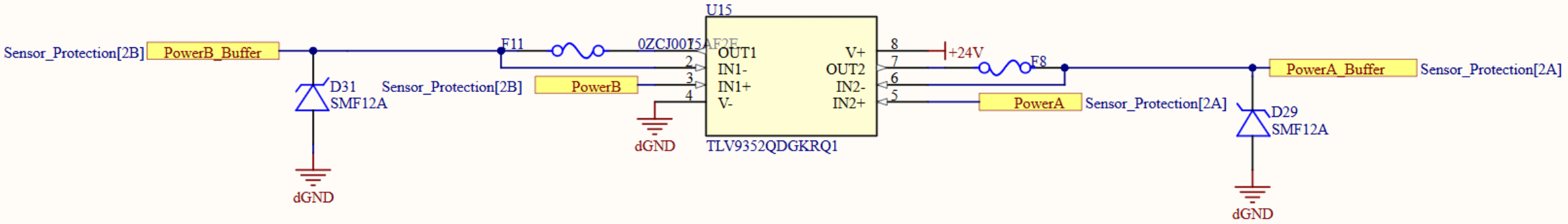
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Trade #3 – Sensor Powering OCP

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2026 Sensor Board V2 Sensor Protection

Problem: Current OCP circuit caused thermal issues

Options:

- Load switches
- PDB (CSA + high side switch w/ updated ltc7004)
- PTC fuses
- Standard fuses

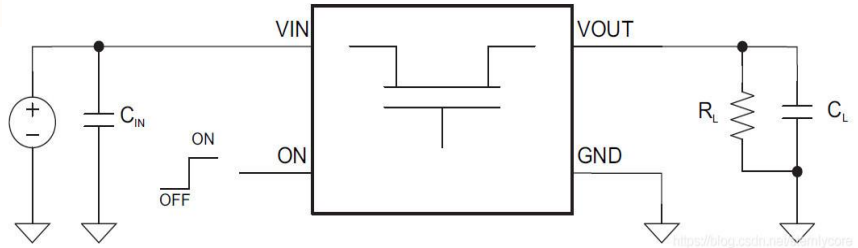
Factors: implementation, size, functionality, and cost

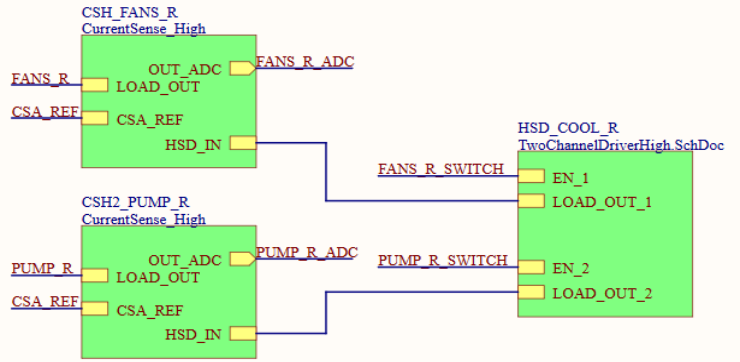
DCR Requirements:

- PFR.2: Sensor board shall provide the specified current and have overcurrent protection (OCP)









Trade #3 – Sensor Powering OCP

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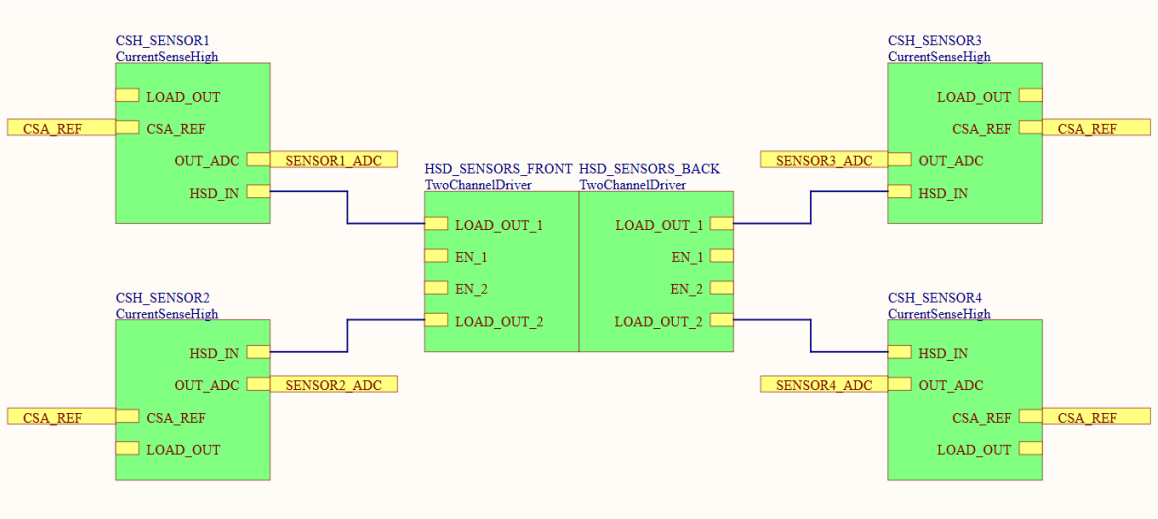
2025-2026 Trade:

- Top choices: PTC fusing, op-amps
- Other options: ORing controllers, load switches

2026-2027 Trade:

- Top choices: standard fusing, PTC fusing, PDB fusing
- Other options: ORing controllers, load switches

2027 Sensor Board V1



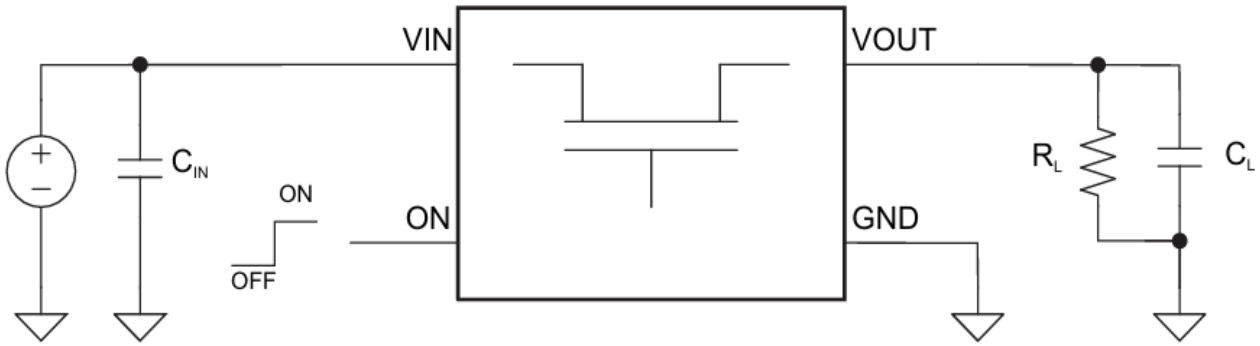


Figure 1. General Load Switch Circuit Diagram

Trade #3: OCP Options

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Category	Option #1: PDB fusing	Option #2: PTC/Polyfuses	Option #3: Ideal Diode Controllers (ORing Controllers)	Option #4: Load-switch IC	Option #5: Regular Fuses
Implementation	Schematics exist already, know what to expect from it	Easier and faster to install	Requires MOSFET selection	Requires more research to decide component, new	Easy to implement, have to replace every time
Size and Complexity	Slightly more complex, takes up more space	Smaller, less complex	Increased circuit complexity due to MOSFET and controller	Covers large area	Small, much less complex
Functionality	More precise trip point, faster response time, configurable OCP, data for debugging	Trip point not as predictable, slower response	Emits less heat than most options, adds reverse polarity protection	Essentially a FET with extra features	Have to replace every time it breaks
Cost	Most expensive	Much cheaper	More expensive	Cheaper	Cheapest

Trade #3: Selection

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Decision: Decided on PTC fusing for offboard OCP due to a variety of reasons:

- Load-switch ICs only offer automatic reset over standard fuses
- PTC fusing is the cheapest, smallest, easiest to implement method, and automatically resets
  - Component selection (current PTC fusing) on next slide
- However, we will run PDB fusing on offboard 5VD and 3V3D for debugging purposes --> Contingent on layout restrictions
  - Offboard sensor harness testing (not analog because shunt voltage drop bad)
  - Tentative kill date Nov 5th

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Trade #4: Layout Changes: Grounding

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- **Problem:** digital switching noise (MCU, CAN, bucks) flows through ground and can corrupt analog sensor readings at the ADC.
- **Last Year:** separate analog ground (AGND) and digital ground (DGND). Option #1
- **DCR:** PFR.4 accurate ADC measurements, PFN. 1 simpler, organized schematics and layout

Option #1: Split AGND / DGND



Separate analog and digital planes, joined at one star point

Option #2: Solid plane, partitioned



One continuous plane; analog and digital parts placed in separate areas

Trade #4: Grounding Options

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Category	Option #1: Split AGND / DGND planes	Option #2: Solid plane, partitioned (chosen)
Noise Isolation	Cuts the plane in two so analog and digital return currents can't mix	Puts analog and digital parts in separate areas; each return current flows under its own trace, so digital current stays out of the analog area
Return Path / EMI	All analog–digital traces must cross at one star point → crowded routing, and their return currents share one narrow path (higher impedance, crosstalk)	Return current flows directly under every trace, wherever it's routed → small current loop, low EMI
Layout Risk	One misrouted trace across the gap creates a large current loop → more EMI; joining the planes at more than one point creates ground loops	Simple rules: group parts by type, keep digital traces out of the analog area
ADC Datasheet (ADS131M08)	Only if a solid plane isn't practical; then join the planes directly at the ADC	Recommended: a dedicated, unbroken ground plane layer for best performance

Trade #4: Grounding Options

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11 Layout

11.1 Layout Guidelines

For best performance, dedicate an entire PCB layer to a ground plane and do not route any other signal traces on this layer. However, depending on restrictions imposed by specific end equipment, a dedicated ground plane may not be practical. If ground plane separation is necessary, make a direct connection of the planes at the ADC. Do not connect individual ground planes at multiple locations because this configuration creates ground loops.

Route digital traces away from all analog inputs and associated components in order to minimize interference.

Use C0G capacitors on the analog inputs. Use ceramic capacitors (for example, X7R grade) for the power-supply decoupling capacitors. High-K capacitors (Y5V) are not recommended. Place the required capacitors as close as possible to the device pins using short, direct traces. For optimum performance, use low-impedance connections on the ground-side connections of the bypass capacitors.

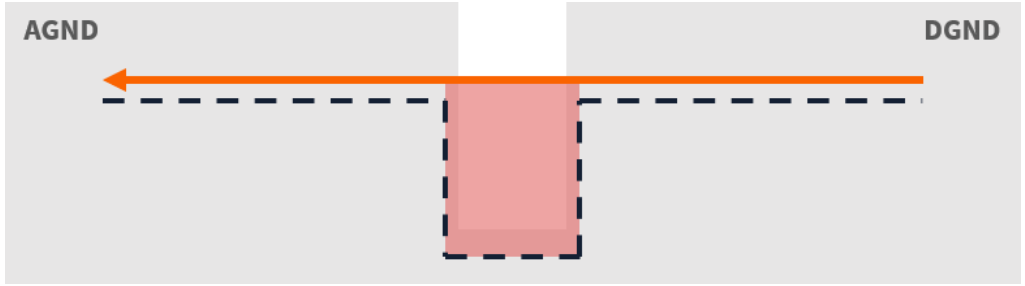
When applying an external clock, be sure the clock is free of overshoot and glitches. A source-termination resistor placed at the clock buffer often helps reduce overshoot. Glitches present on the clock input can lead to noise within the conversion data.

Grounding: Selection

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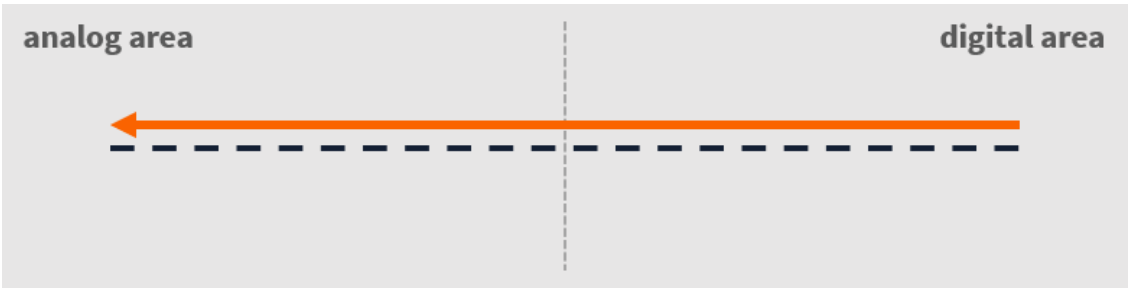
Decision: one solid ground plane (single GND net), partitioned layout

Split AGND / DGND



Return can't cross the gap, so it detours to the star point → large loop (red), more EMI

Solid plane, partitioned (chosen)



Return current stays right under its trace → small loop

signal return current loop area

Why

- Separation comes from where return currents flow, not from cutting copper
- A split adds failure risk: any trace across the gap creates a large current loop
- ADS131M08 datasheet: a dedicated, unbroken ground layer gives best performance

Implementation

- Dedicated ground layer, no signal traces on it
- Analog parts in one region; MCU, CAN, bucks, and other digital/switching parts in another
- Digital ICs placed away from analog ICs
- Fast digital traces kept away from sensitive analog traces
- No digital traces through the analog region
- ADC ground pins straight to the plane; decoupling caps at every power pin

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- Board Purpose
- Roses/Thorns
- Current Board Overview
- ADC Selection
- Analog Signal Filtering Chain
- Sensor Powering OCP
- Layout Changes
- MCU Pinout
- Software
- Conclusion

MCU Pinout

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- Mostly carrying over from last year
- Sensor digital I/O subject to change
- Changes:
  - More board monitoring analog inputs
  - microSD slot/SDMMC peripheral

Category	Peripherals	Description
CAN	FDCAN (3)	CAN 1, 2; private sensor CAN
External ADC	SPI (1), TIM (1, 1 ch.)	ADS131M08 communication and 8.192 MHz clock
Digital sensor communication	I2C (3)	VL53L4CD time-of-flight
	SPI (1)	Corner accelerometer
Digital inputs	TIM (1, 2 ch.)	Flow rate sensors
	TIM (1, 2 ch.)	Lap trigger, GPS PPS (timebase)
NVM	OCTOSPI (1)	QSPI NOR Flash
	SDMMC (1)	microSD card slot
Board monitoring	ADC (<= 17 ch.)	Power rail voltage monitoring, board temps, offboard power current sense

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Software Overview

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Task	Description	Period	Priority
System	Init and config/external command handling, watchdog Board and software status monitoring, offboard power management	100Hz	2
ADC	Collect analog sensor data from the ADS131M08, apply board HW calibrations Minimum configurable ADC sampling rate is 1kHz, task receives multiple samples between steps	1kHz	4
TOF	One instance per sensor (expected 3). Handles I2C communication and data processing for VL53L4CD TOF sensors. Significant compute overhead	100Hz	3
Accelerometer	One instance per sensor (expected 1). Handles batched high-frequency SPI accelerometer data	<= 1kHz	4
CAN TX	CAN transmit tasks (one per bus)	Event	4
CAN RX	CAN receive task. Low priority for car-side buses, highest priority for private CAN	Event	1, 5
NVM	microSD sensor data logging (config-dependent)	Variable (I/O)	1

Software Overview

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New:

- Data manager: killed the single high-priority task, now a module
  - Clean separation between sensor data acquisition and post-processing/logging
- Module hierarchy
  - Separate all code that touches hardware from application code
  - Mocks for the full hardware layer -> SIL and unit testing
- System manager task
  - Board, sensor, and software diagnostics/logging
- Configuration
  - DAG -> fixed list of post-processing steps that can be configured for each sensor

NVM

128Mb SOIC-8 QSPI Flash (standard):

- Configurations, calibrations
- System parameter/application data storage
- Diagnostic logging

microSD card:

- Optional time-synced bulk data storage
- Allows the board to be used for standalone logging
- Real-world write performance ~10-12 MB/s
- Kill if layout is tight

Signal Processing

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- Most sensors capable of higher sampling rates than logging bandwidth supports
- All analog channels sampled at the same frequency (synchronous sampling), min. 1kHz
- Need to reduce data rate before logging

Oversample-and-average vs. FIR LPF/decimate:

- Oversampling == FIR LPF with N equally-spaced taps
- Correctly-designed LPF & decimation has a better frequency response and properly attenuates out-of-band frequencies
  - Preferred, but significantly more compute overhead. Will be combined with oversampling if needed

Timing and Jitter

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Conclusion

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Main Changes

- ADC sticking with ADS131M08
- Signal buffering thermistor ADC side buffer removed
- OCP is copying PDB --> Will be killed if layout is chopped and use just PTCs instead (speak to dex ab fusing)
- Ground layers changed from separate digital and analog grounds to continuous ground

Next Steps/Timeline:

- Waiting on information from LV enclosures for confirmation about connectors, making next steps based off assumptions for what will go on snsb
- Schematics started
  - Will delegate the rest soon and plan to finish by Oct 28th (Schematic review on the 30th)
  - Layout will start soon after schematics are done (FDR is Nov. 12 when layout and schematics need to be done) kill date for OCP will be Nov. 5th
  - Continue following cd and sw timelines